ZoneCast

User Manual

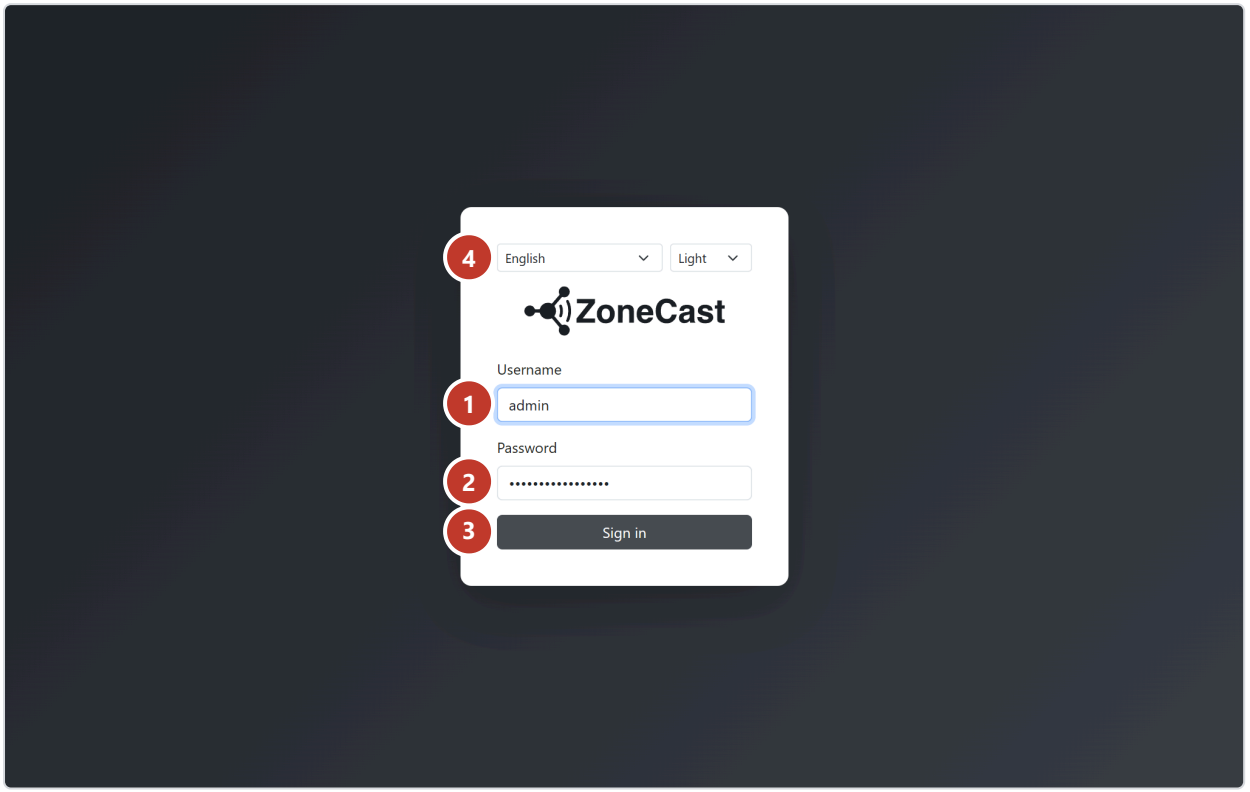
Illustrated guide to the dashboard's features

Application version 1.6.4 — November 2026

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1. Login



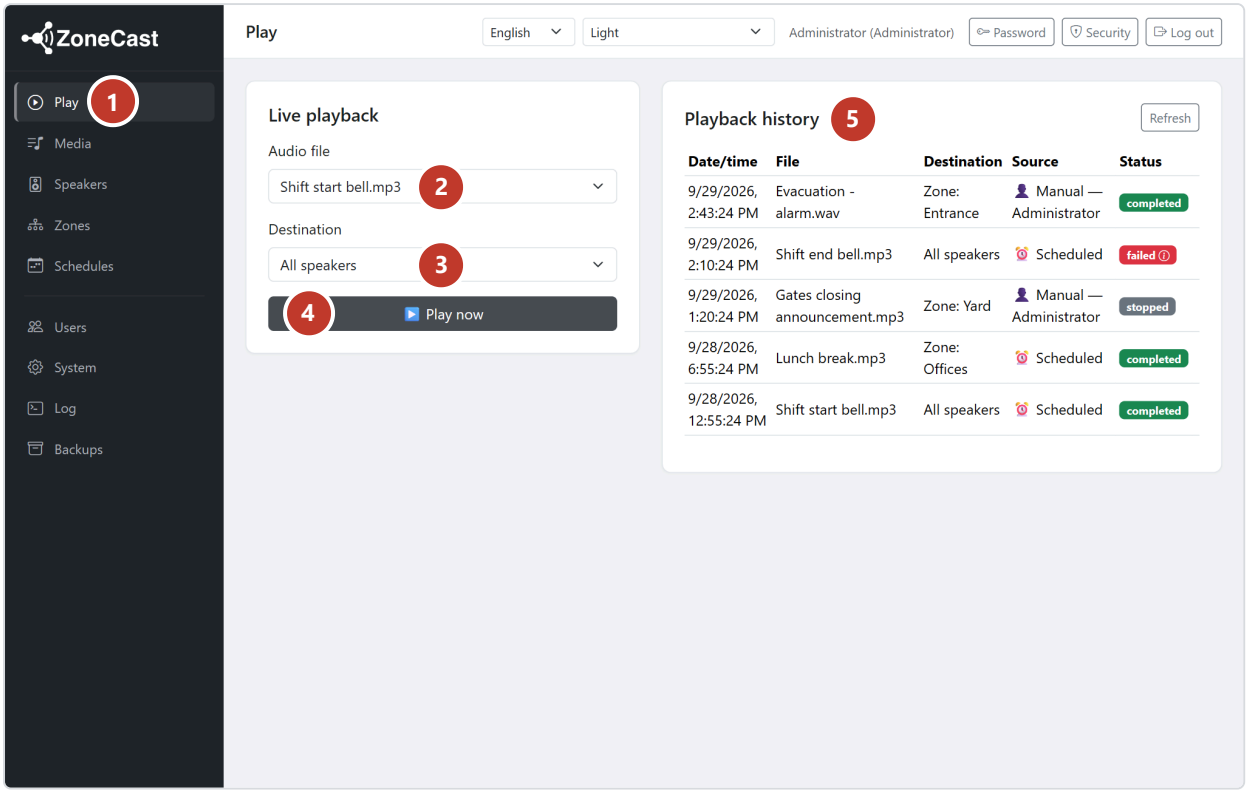
The login page is reachable at the server's address (e.g. `http://192.168.187.10`).

1. **Username** field — the username assigned by the administrator.
2. **Password** field.
3. **Sign in** button — if the account has 2FA enabled, after submitting you're also asked for the 6-digit authenticator app code (or a recovery code).
4. **Language** and **theme** (light/dark/automatic) pickers — see section 15.

> After 5 failed attempts within 5 minutes with the same username from the same computer (IP address), that username can't sign in from that computer for up to 5 minutes, even with the right password: the page shows "Too many failed attempts. Try again in N seconds." Signing in from another computer still works. The same limit applies, separately, to wrong 2FA or recovery codes.

> A session expires after 12 hours without use (the page's own automatic refreshes don't count): the next action then takes you back to the login page. Closing the browser doesn't sign you out: on a shared PC always use **Log out**.

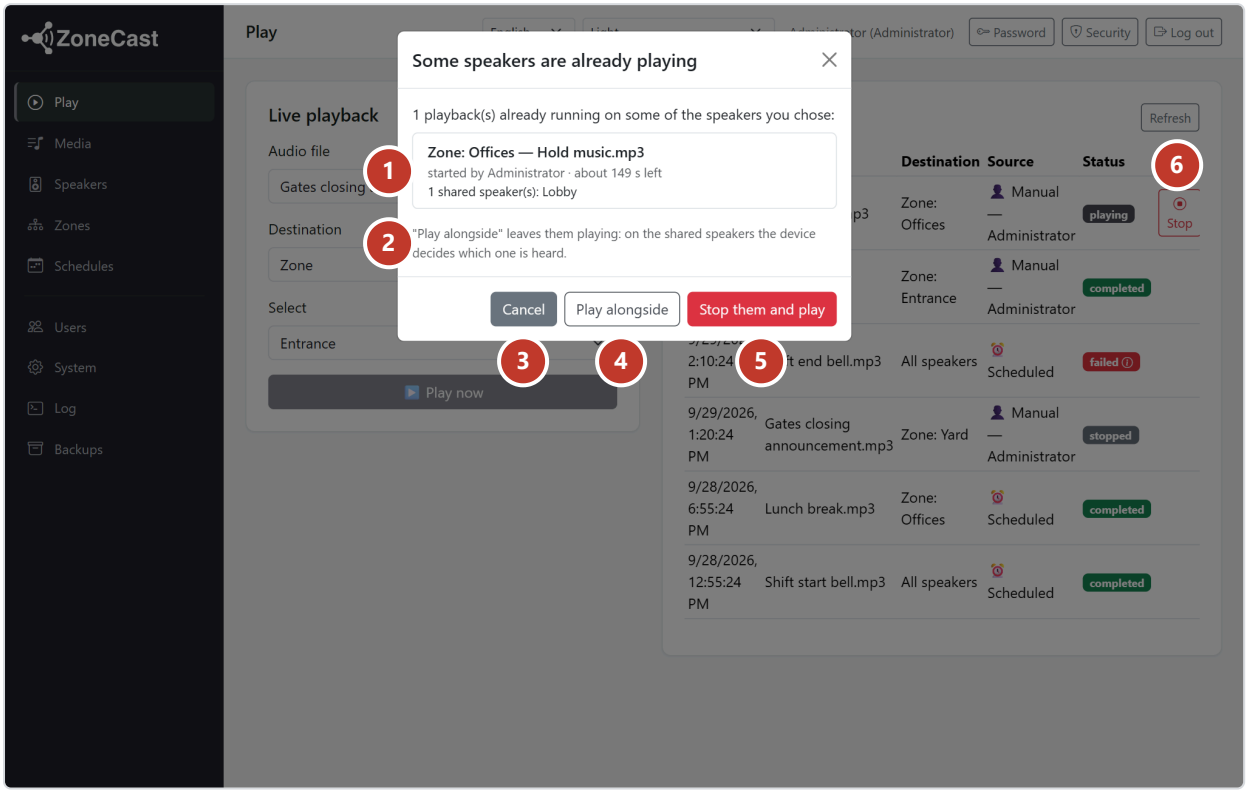
2. Live playback



This is the landing screen: it lets you start an immediate announcement to one or more speakers.

1. Sidebar menu — switch between dashboard sections.
2. **Audio file** — choose among the files already uploaded in Media.
3. **Destination** — "All speakers", a zone or a single speaker; for a zone or a speaker a **Select** list appears below to pick which one.
4. **Play now** button — starts multicast playback immediately. If some of the chosen speakers are already playing, you're asked what to do first (see section 3).
5. **Playback history** — the latest 30 playbacks, newest first: date/time, file, destination, source (Manual with the user's name, or Scheduled) and outcome: completed, stopped, failed (the ⓘ symbol shows the reason). A playback still running shows a **Stop** button. *Completed* means the audio was sent over the network, not that every speaker played it: a device that is off or **Not in sync** misses it with no error.

3. Live announcement on busy speakers

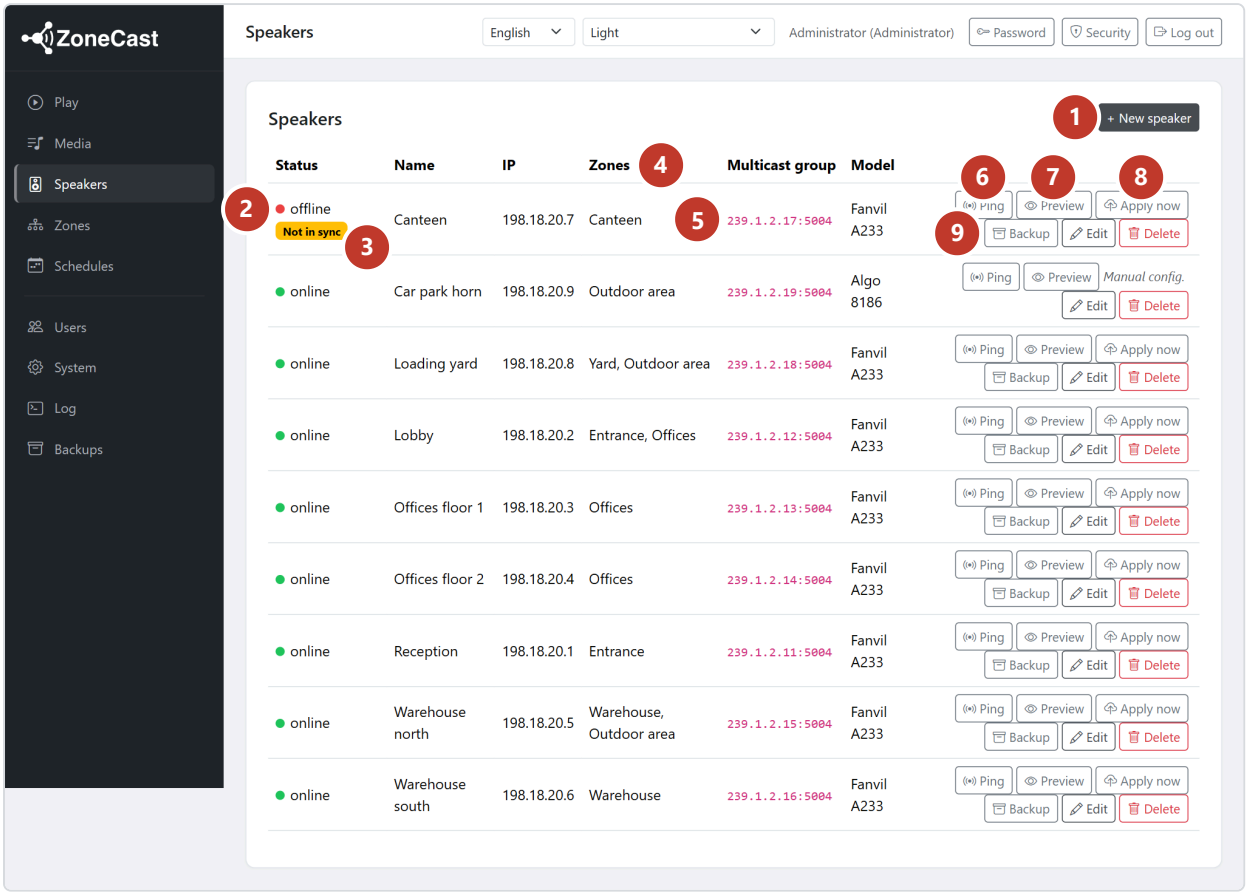


If some of the speakers you chose are already playing (another announcement or a schedule), **Play now** doesn't cut in straight away: a dialog shows what is playing and lets you decide.

1. What is already playing — destination and file, who started it (or which schedule), about how many seconds are left, and the **shared speakers**. A playback on exactly the same destination is marked: it is stopped whichever way you start yours, "Play alongside" included; only Cancel leaves it playing. If every listed playback is on the same destination, "Play alongside" and its reminder are not shown.
2. Reminder of what "Play alongside" does.
3. **Cancel** (default) — nothing is started.
4. **Play alongside** — leaves the other playback running; on the shared speakers the device decides which one is heard.
5. **Stop them and play** — stops the listed playbacks and starts yours.
6. **Stop** — stops a running playback from the history.

> The dialog also lists (up to 5) the schedules due on the same speakers while your announcement plays: a bell **waits** for a live announcement to end (up to 60 s past its time) and then plays; if the announcement lasts longer, the dialog says it won't play (it is recorded as failed in the history). The rule applies to every live announcement, but the list only appears in this dialog: if nothing is playing on the chosen speakers, the announcement starts at once with no warning about upcoming bells.

4. Speakers — list and actions

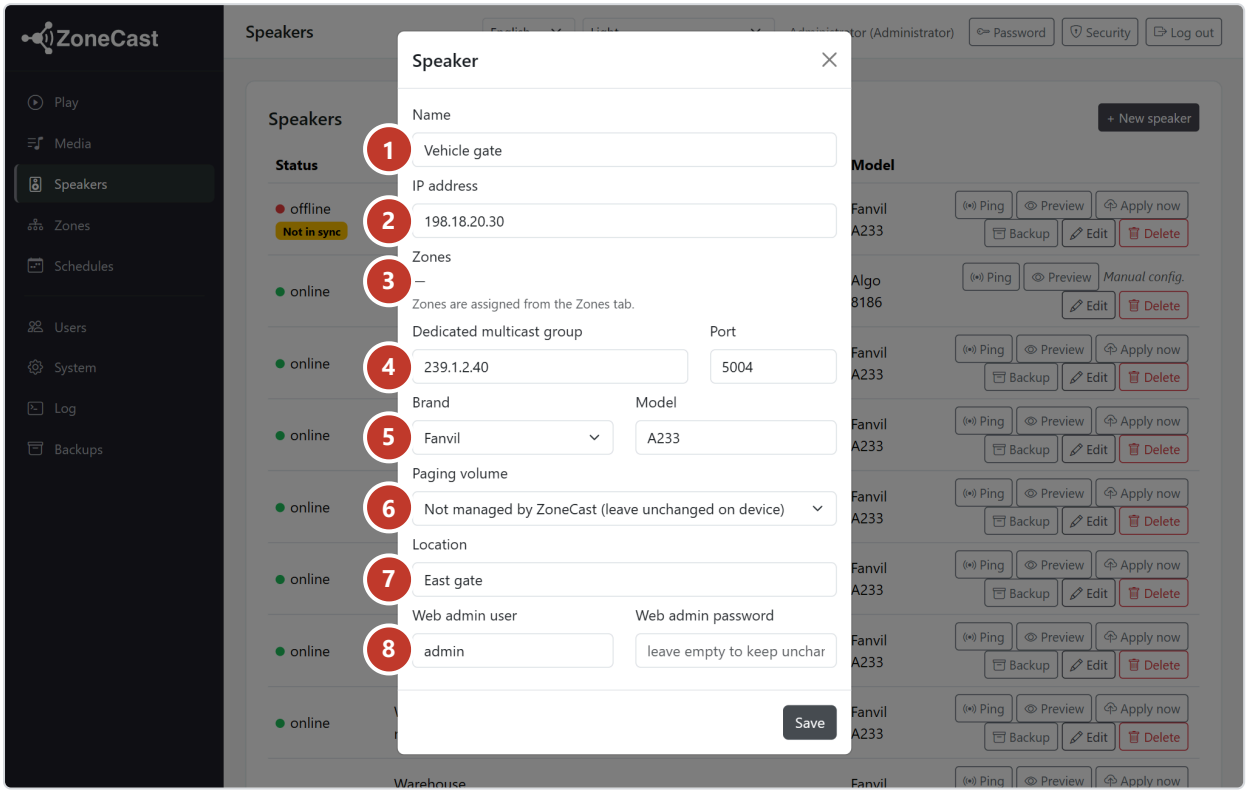


List of all registered speakers with their online/offline status (checked by the server every 5 minutes; the table shows the status read when the page was opened — reload the page or press Ping to update it).

1. **+ New speaker** — opens the creation form (see section 5).
2. **Status** dot — green online, red offline, grey "unknown" (newly added, not checked yet).
3. Zone setup badges: **Updating...** while the new zone setup is being sent to the device; **Not in sync** if the device didn't accept it. If it was offline, it is re-sent automatically at the first periodic check (every 5 minutes) where the device answers; if the credentials are wrong, correct them with **Edit** (saving re-sends it at once) or use **Apply now**. Admins also get a login alert. Until then, the device keeps playing according to its previous setup.
4. **Zones** — all the zones the speaker belongs to (a speaker can be in several zones).
5. **Multicast group** assigned to that speaker (dedicated address:port).
6. **Ping** — checks the device's reachability right away. "Online" means its web page (HTTP, port 80) answers: credentials and audio are not checked.
7. **Preview** — shows what would be sent to the device without actually applying it.
8. **Apply now** — writes the multicast configuration to the device (supported brands only, currently Fanvil; otherwise "Manual config." is shown).
9. **Backup / Edit / Delete** — back up the device's configuration, edit its data, delete it: refused if the speaker is the destination of a schedule (the message lists the schedules to change first). Backups remain available in the archive even after deletion. **Backup** opens the speaker's backups: **+ New backup** downloads the configuration from the device now; ZoneCast doesn't restore it (load it from the device's web page).

> Operators see only Ping, Preview, Edit and Delete: **Apply now**, **Backup** and "Manual config." are for administrators.

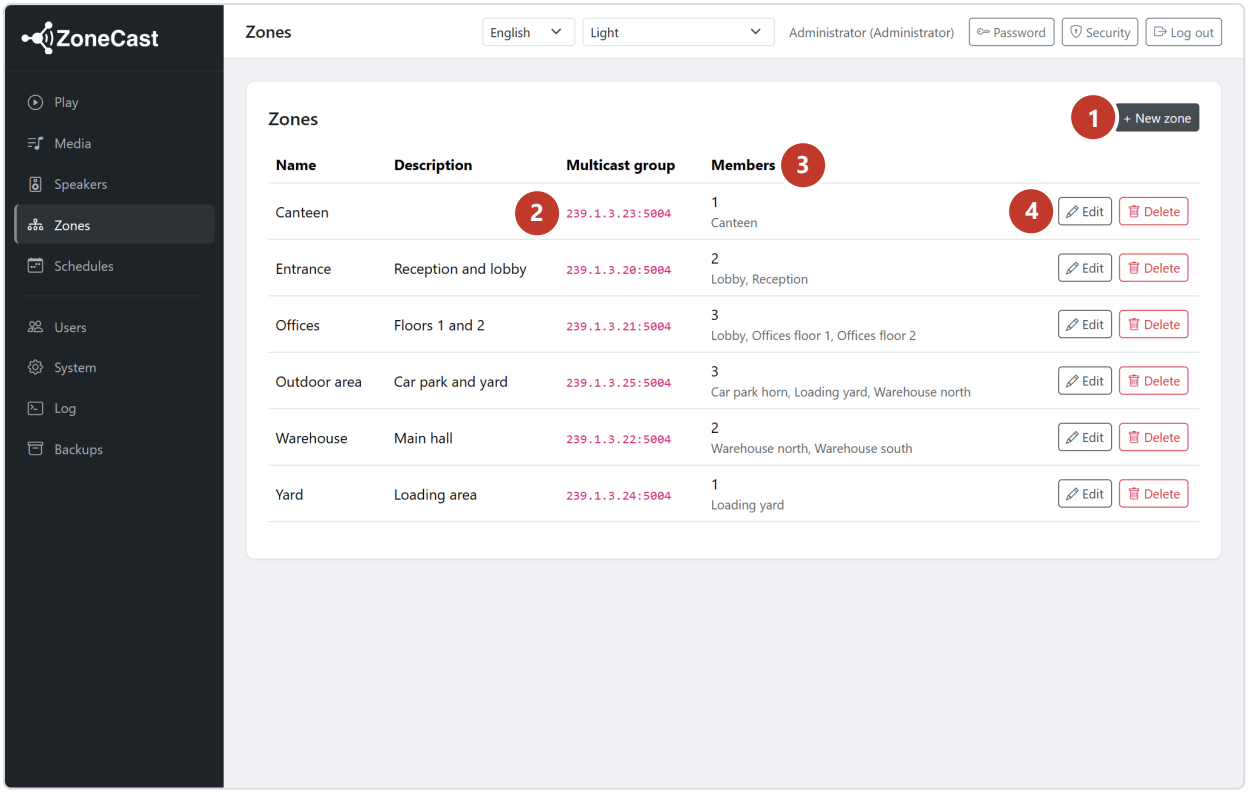
5. Adding/editing a speaker



1. **Name** — the label shown everywhere in the dashboard.
2. **IP address** of the device on the LAN.
3. **Zones** — read-only: the zones this speaker is in ("(none)" if it isn't in any; "—" for a new speaker). Zones are assigned from the Zones tab (see section 7).
4. **Dedicated multicast group** and **Port** — used to reach this single speaker. Each group (address:port) can belong to only one speaker or one zone and can't be the "All speakers" one (default 239.1.1.99:5004); the IP address must be unique too, otherwise saving is refused.
5. **Brand** — dropdown with the most common multicast brands (Fanvil, Algo, CyberData, Grandstream, Axis, Tiptel, Akuvox), "(unspecified)", preselected for a new speaker, and **Other**, which lets you type a name freely; **Model** is free text. In this version automatic actions exist for **Fanvil** only: only with that brand do the Paging volume field and, for administrators, Apply now/Backup appear, and is the multicast list updated on the device automatically. Configure other brands by hand following the Preview.
6. **Paging volume** (supported brands only) — "Not managed by ZoneCast" leaves the device's setting unchanged; otherwise choose "Device default" or a level from 1 to 9.
7. **Location** — where the speaker is; it also helps find it in the zone member list.
8. **Web admin user/password** — used by the app to apply configurations; the password is encrypted at rest in the database (leave it empty to keep the saved one; type it again when changing the IP address or port).

> On a Fanvil, ZoneCast rewrites the device's whole multicast list (never SIP or network settings) when you save the speaker, when its zones change and with **Apply now**: groups entered by hand on the device's web page are erased. **Deleting** a reachable Fanvil speaker empties the device's list too, so it stops playing announcements. If the device doesn't answer, or for other brands, a message warns that it will keep playing until you reconfigure it from its web page.

6. Zones



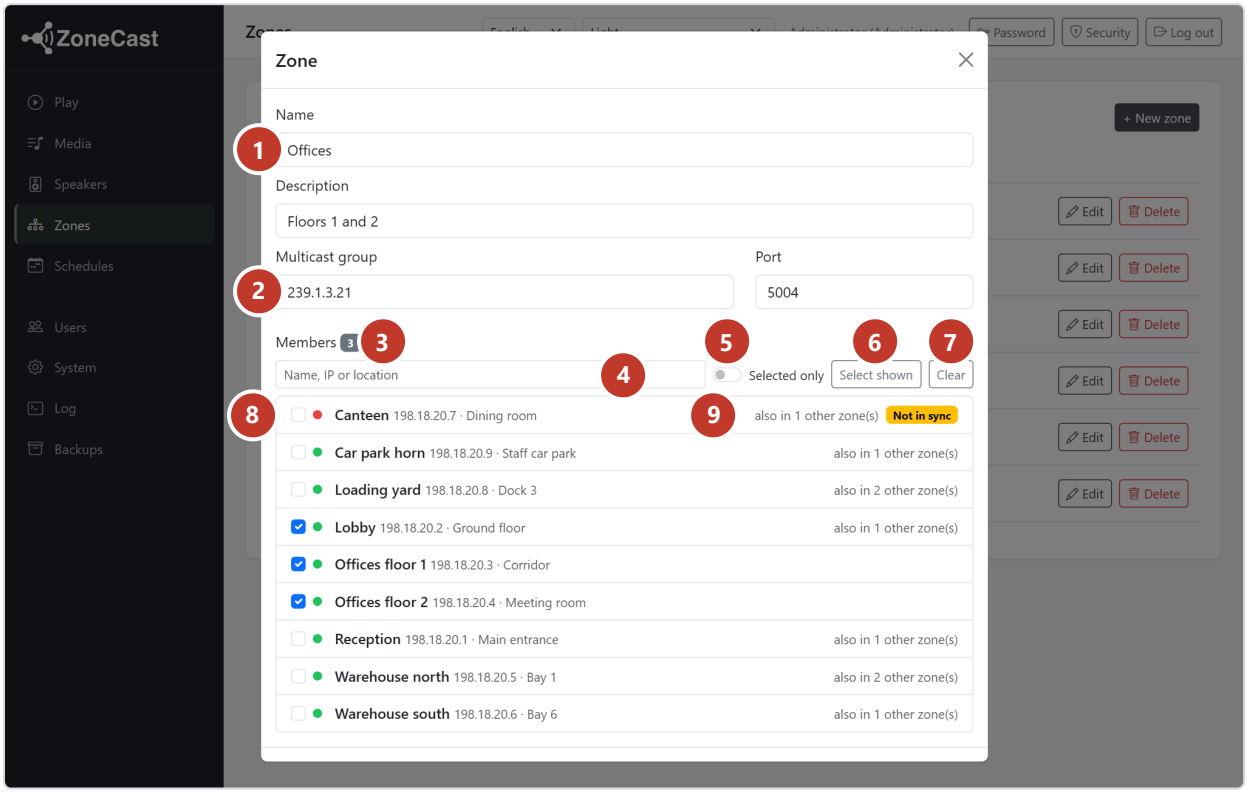
Zones group multiple speakers together (e.g. a floor, a department) so you can reach them together from Playback or from Schedules. The same speaker can be in several zones (e.g. "Lobby" is in both Entrance and Offices).

1. **+ New zone** — name, description, zone multicast group and members (see section 7).
2. **Multicast group** of the zone (address:port).
3. **Members** — how many speakers belong to the zone, with the names of the first three underneath ("+N" when there are more).
4. **Edit / Delete** — changing a zone's members, address or name, or deleting it, updates in the background the affected devices that support it (currently Fanvil only): the Speakers tab shows "Updating…" and then, if the device doesn't accept it, "Not in sync". Speakers of other brands ("Manual config.") must be reconfigured by hand following their Preview.

> Click a column header to sort A→Z, click again for Z→A (a ▲ / ▼ appears next to the title), in the Speakers, Zones, Media and Schedules tables.

> A zone, speaker or audio file used by a schedule can't be deleted: the message lists the schedules to change first.

7. Zone members

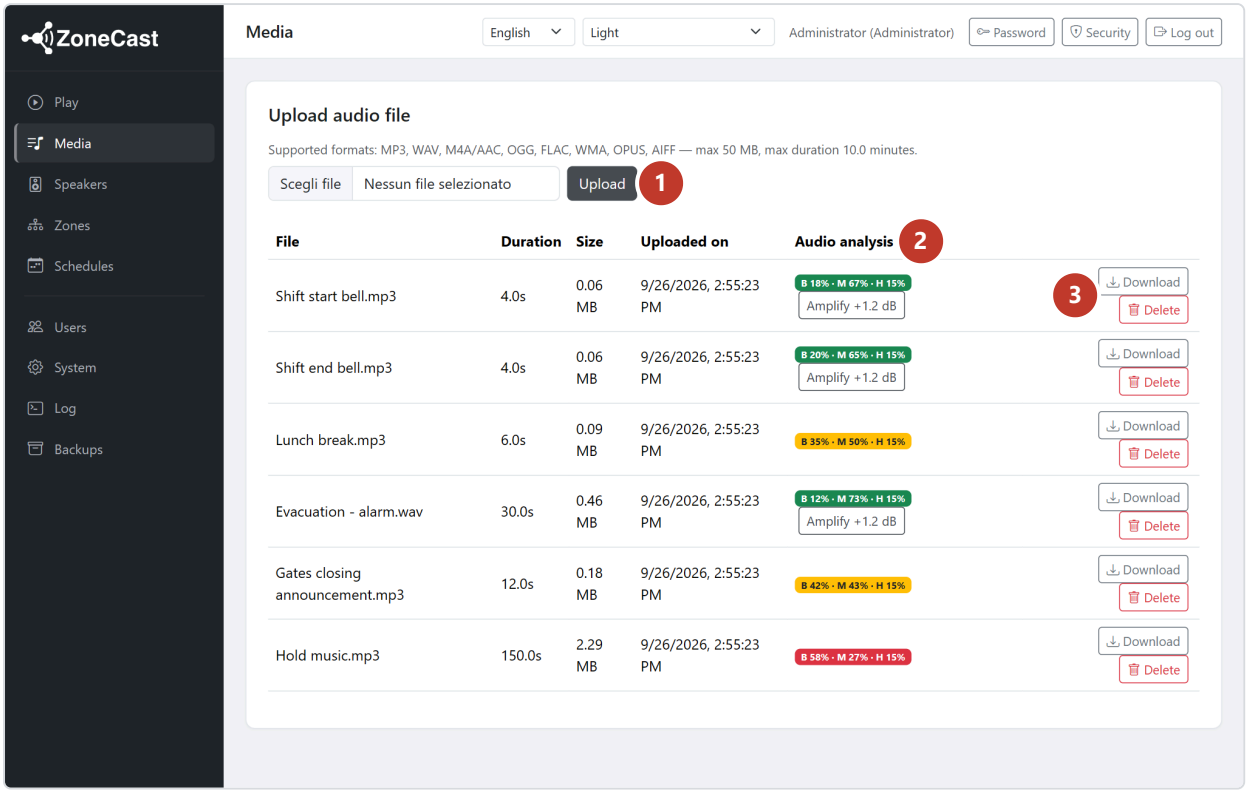


A zone's speakers are chosen in the zone form (**+ New zone** or **Edit**), then saved with **Save** at the bottom.

1. **Name** and **Description**.
2. **Multicast group** and **Port** of the zone.
3. Member counter — how many speakers are selected.
4. Search box — filters the list by name, IP or location.
5. **Selected only** — shows only the ticked speakers.
6. **Select shown** — ticks every speaker currently shown (e.g. after a search), except those whose device has already reached its zone limit.
7. **Clear** — unticks all.
8. One row per speaker: tick box, status dot, name, IP and location.
9. How many **other zones** the speaker is already in, and its "Updating…" or "Not in sync" badge if any. A speaker whose device has reached its zone limit (18 zones on a Fanvil) is greyed out and can't be ticked.

> Tables are loaded when the page opens and only refresh after your own changes. If someone else may have changed the same zone, schedule or speaker, reload the page before editing it: **Save** rewrites every field, the member list included.

8. Media — upload and audio analysis



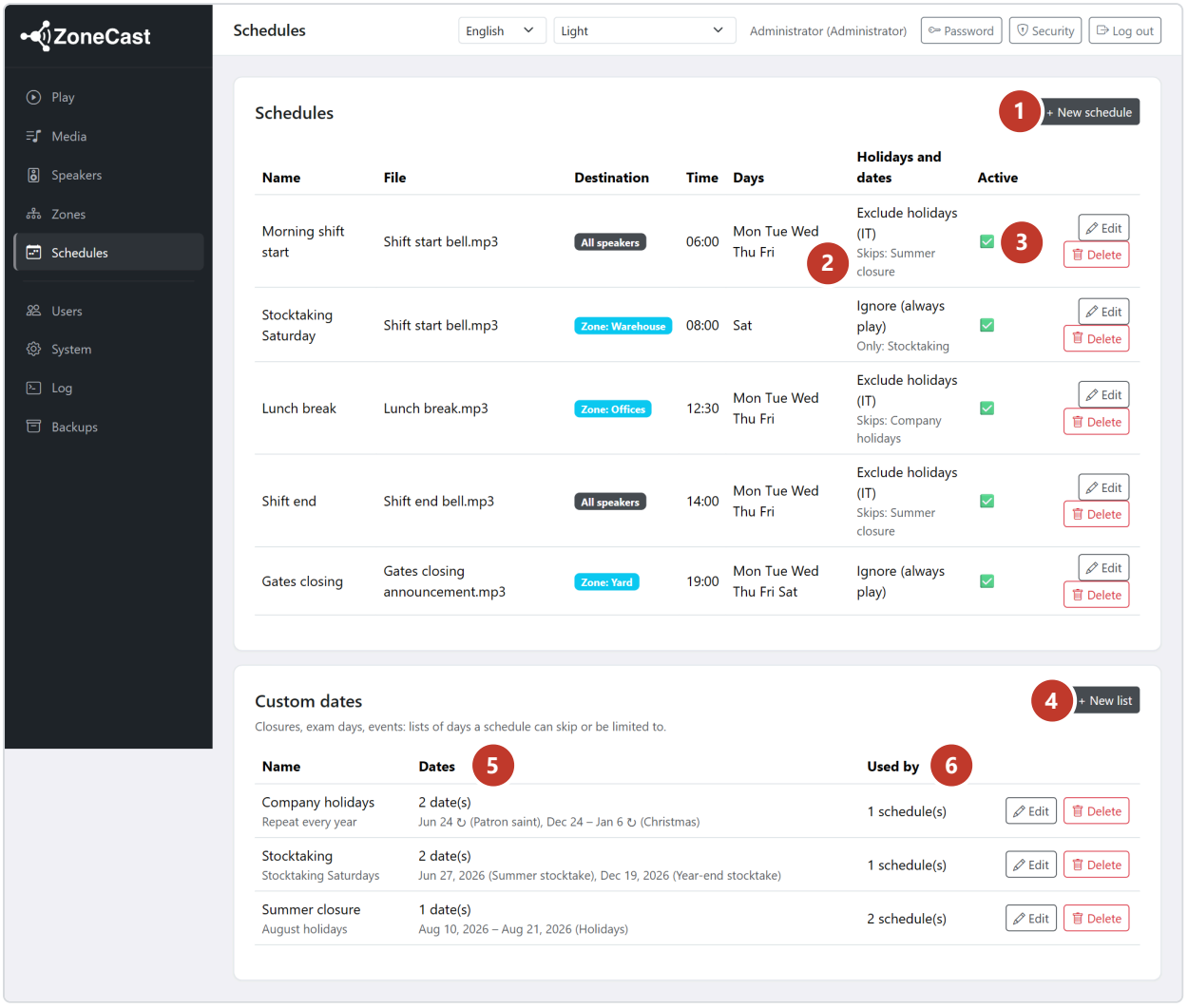
Every uploaded file is automatically analyzed and assessed for playback on PA horns (which reproduce low frequencies poorly).

1. Pick a file and press **Upload** — MP3, WAV, M4A/AAC, OGG, FLAC, WMA, OPUS, AIFF (max 50 MB, max 10 minutes), automatically converted for playback.
2. **Audio analysis** badge — share of bass (B), mid (M) and high (H) frequencies, colored green/yellow/red depending on how dominant the bass is: hovering shows a tooltip with the details. If an **"Amplify +N dB"** button also appears, the file has headroom to be brought up to maximum volume without distortion. One click applies the gain at once, without confirmation, to the stored file itself (Download then returns the amplified version), and it can't be undone; the button is then replaced by a grey **Amplified** badge.
3. **Download / Delete** the file. A file used by a schedule can't be deleted: the message lists the schedules to change or delete first.

> A yellow or red badge signals that the audio might sound weak or unclear on PA horns: it doesn't prevent playback, it's just a hint. Playback is in telephone quality (mono, 8 kHz, G.711): clear voices and simple chimes work best.

> A file can't be replaced: uploading one with the same name adds a second entry and schedules keep using the old one. Upload the new one under a different name, pick it in the schedules, then delete the old one.

9. Schedules, holidays and custom dates



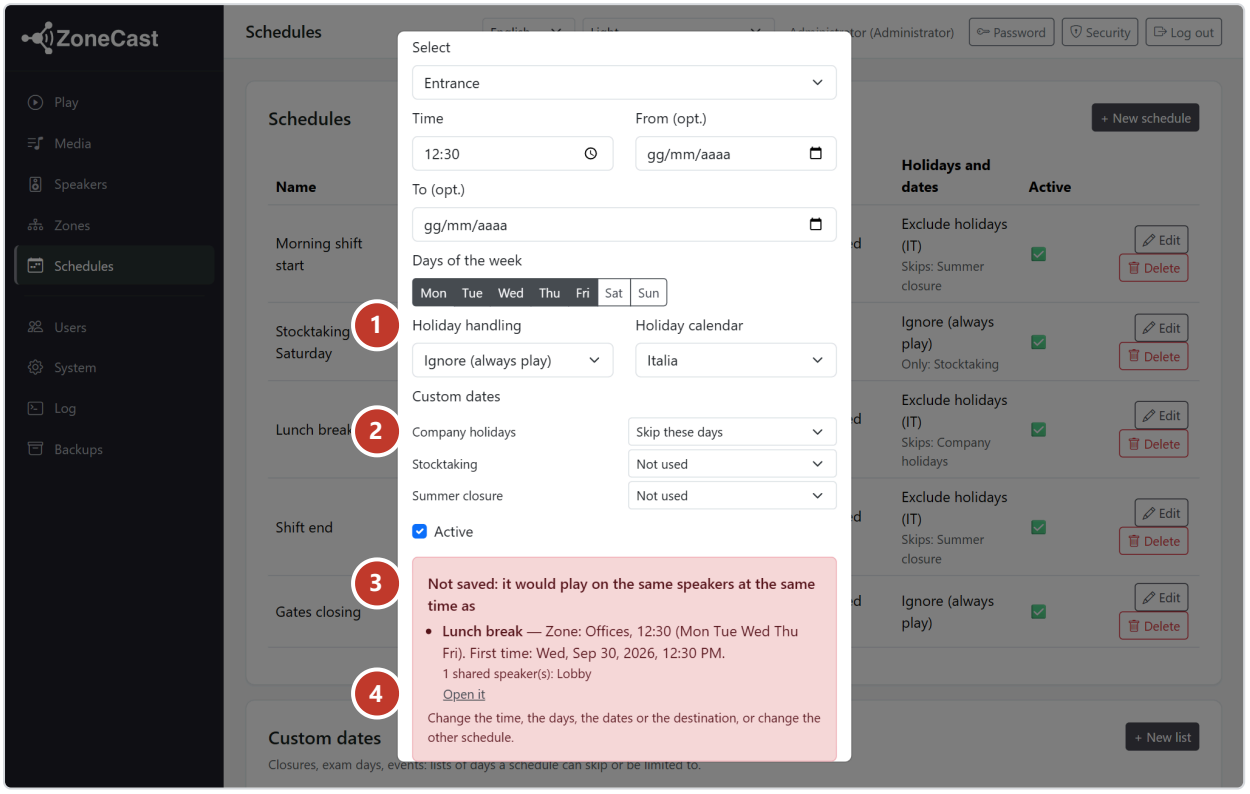
Plan automatic playback of a file at recurring times and days. Every schedule has a name, a media file, a destination (all speakers/zone/speaker), a time, the days of the week and an optional date range.

1. **+ New schedule** — opens the schedule form (see section 10).
2. **Holidays and dates** — the holiday rule with its country (e.g. "Exclude holidays (IT)", "Ignore (always play)") and, underneath, the custom dates used: **"Skips: …"** or **"Only: …"**.
3. **Active** — an inactive schedule is kept but doesn't play.
4. **+ New list** — creates a list of custom dates (see section 11).
5. **Dates** — number of dates in each list and the first three of them (the rest are shown as "+N"; open **Edit** to see them all); ↻ = every year.
6. **Used by** — how many schedules use the list.

A **warning triangle** next to a schedule means it already plays on the same speakers at the same time as another one (e.g. schedules that already overlapped before the upgrade to 1.6.0, or after a zone's members or a list of dates changed): hover it to see which. Admins also get a login alert. While it lasts, at run time the schedule that starts later stops the other one if they share the destination; otherwise both play.

A bell due while a live announcement is playing on some of the same speakers waits for it to end and then plays; if the announcement is still going after 60 s, the bell is not played: it shows as a failed playback in the history and, for admins, in the login alerts.

10. Overlapping schedule refused



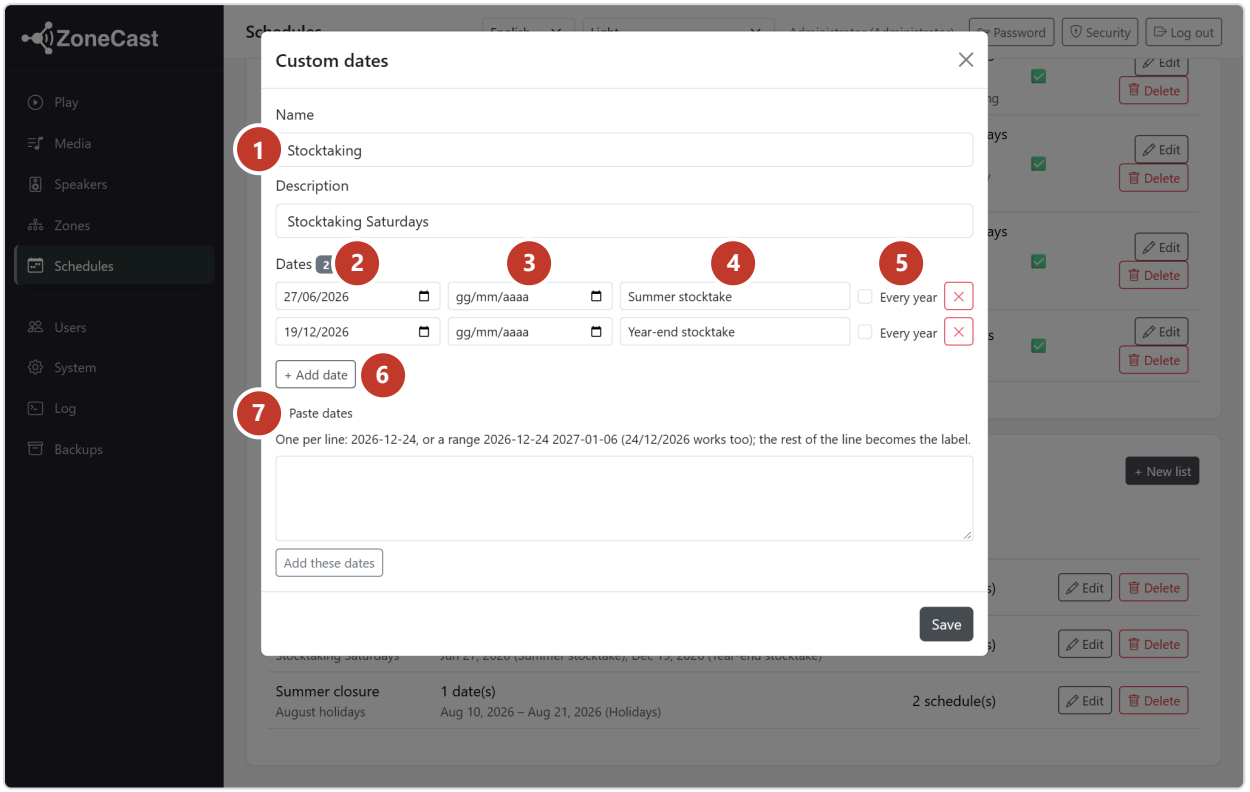
The lower part of the schedule form: time, optional From/To dates, days of the week, holidays, custom dates and **Active**.

1. **Holiday handling** and **Holiday calendar** — pick a country (Italy, United States, United Kingdom, France, Germany, Spain, Japan, China — the menu shows each in its own language, e.g. Deutschland, España) and whether to exclude its holidays, ignore them (always play) or play *only* on holidays.
2. **Custom dates** — one selector per list: **Not used**, **Skip these days** or **Only on these days**.
3. A schedule lasts from its start time until its audio file ends. If it would be playing at the same time as another active schedule on at least one shared speaker, **it is not saved**: the box shows the schedule in the way (destination, time, days), **when they would first clash** and the **shared speakers**.
4. **Open it** — opens the other schedule, to change it instead.

> To fix it, change the time, the days, the dates or the destination — or change the other schedule. The check also counts speakers shared through zones and "All speakers". It applies only when **Active** is ticked: a paused schedule is saved anyway and is checked when it is made active again. Overlaps that already exist don't block unrelated edits (e.g. a rename).

> "Only on these days" and "Only on holidays" narrow the schedule and never add days: the day must also be one of the selected days of the week and inside the From/To range. With several "Only" lists, one match is enough; a day in a "Skip" list is always skipped.

11. Custom dates

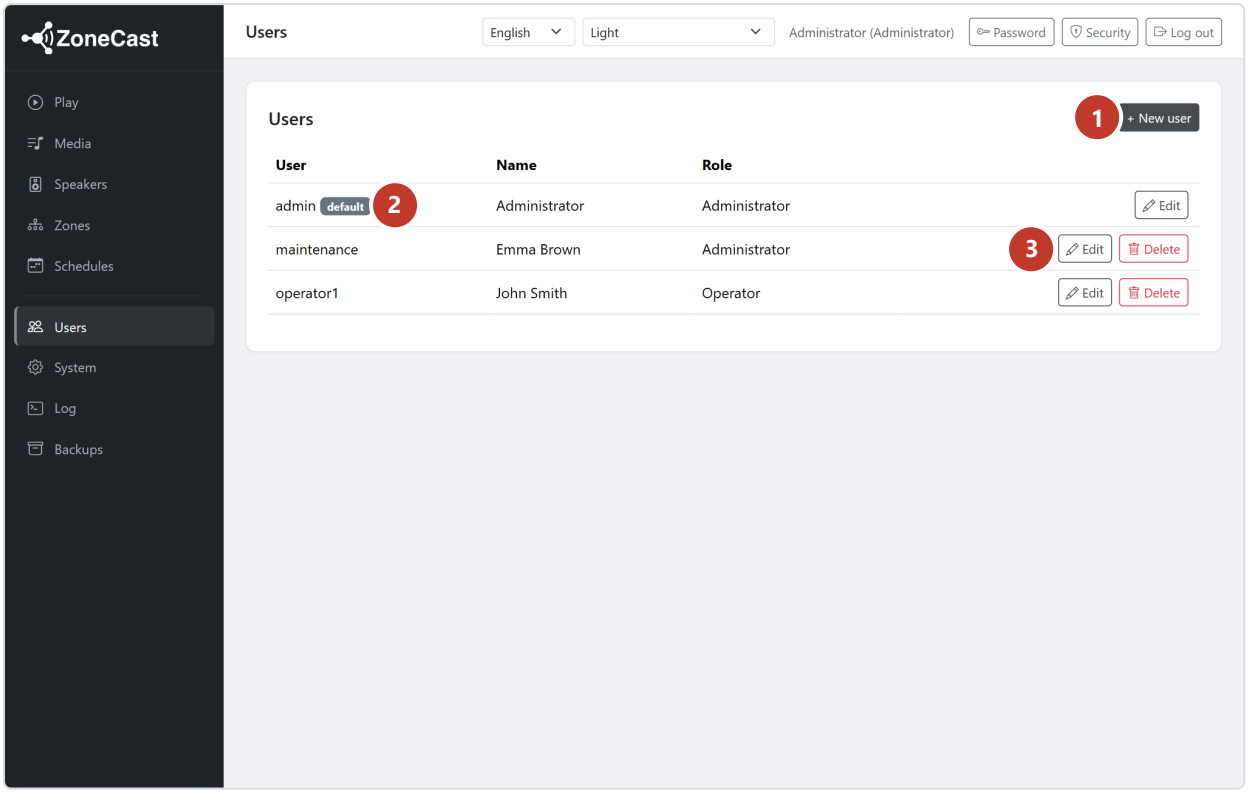


Named lists of days (closures, exam days, events) that a schedule can skip or be limited to. Open with **+ New list** or **Edit** under the schedules list.

1. **Name** and **Description** of the list.
2. First day of the date (or range).
3. Last day — optional, only for a range.
4. **Label** — e.g. "Christmas".
5. **Every year** — repeats the date each year; a yearly range may cross New Year: enter the last day in the following year (e.g. Dec 24, 2026 – Jan 6, 2027). The red ✕ removes the row.
6. **+ Add date** — adds an empty row.
7. **Paste dates** — click it to open the box; one date or range per line (`2026-12-24` , `24/12/2026` , or `2026-12-24 2027-01-06 Christmas` : the rest of the line becomes the label), then **Add these dates** and finally **Save**. Lines not understood stay in the box and are flagged. Pasted dates arrive with **Every year** unticked: tick it where needed.

> A list used by schedules can't be deleted: remove it from them first. Editing a list may make two schedules overlap: this is reported after saving and flagged in the schedules list.

12. Users

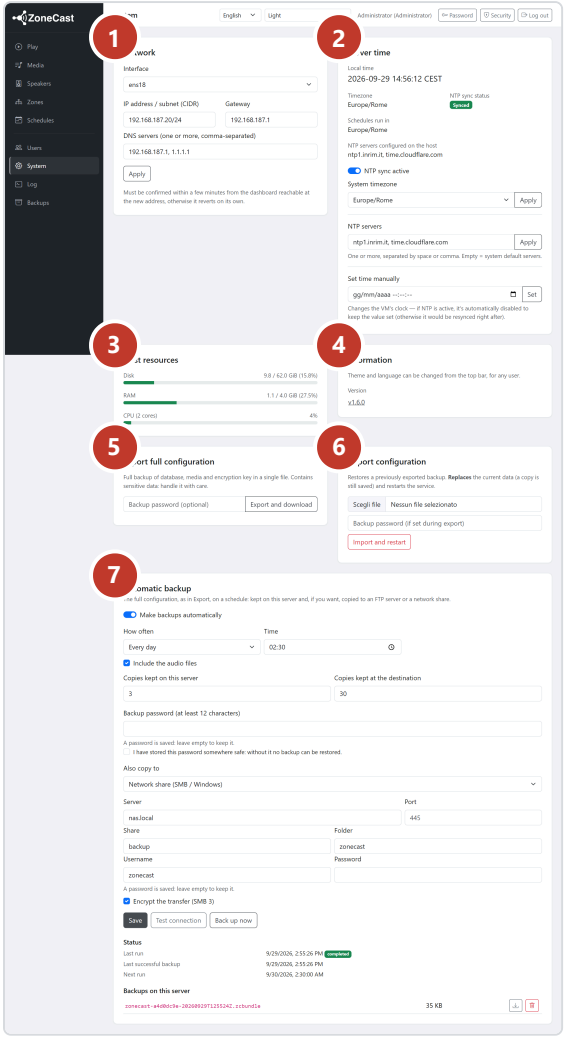


Only administrators can see and manage this section.

1. **+ New user** — available roles: **Administrator** (full access, including the Users/System/Log/Backups sections) and **Operator** (playback, managing speakers/zones/media/schedules).
2. The **default** admin user created on first startup can't be deleted and always stays an **Administrator** (its role is locked in **Edit**): there is always an account to sign in with and manage users and the system.
3. **Edit / Delete** — an administrator can also reset another user's password from here, but can't delete their own account. Passwords must be at least 8 characters; a username can't be changed after creation. After a password change, that user's other sessions are signed out.

Every user can change their own password (**Password** button at the top) and enable 2FA on their own (see section 18).

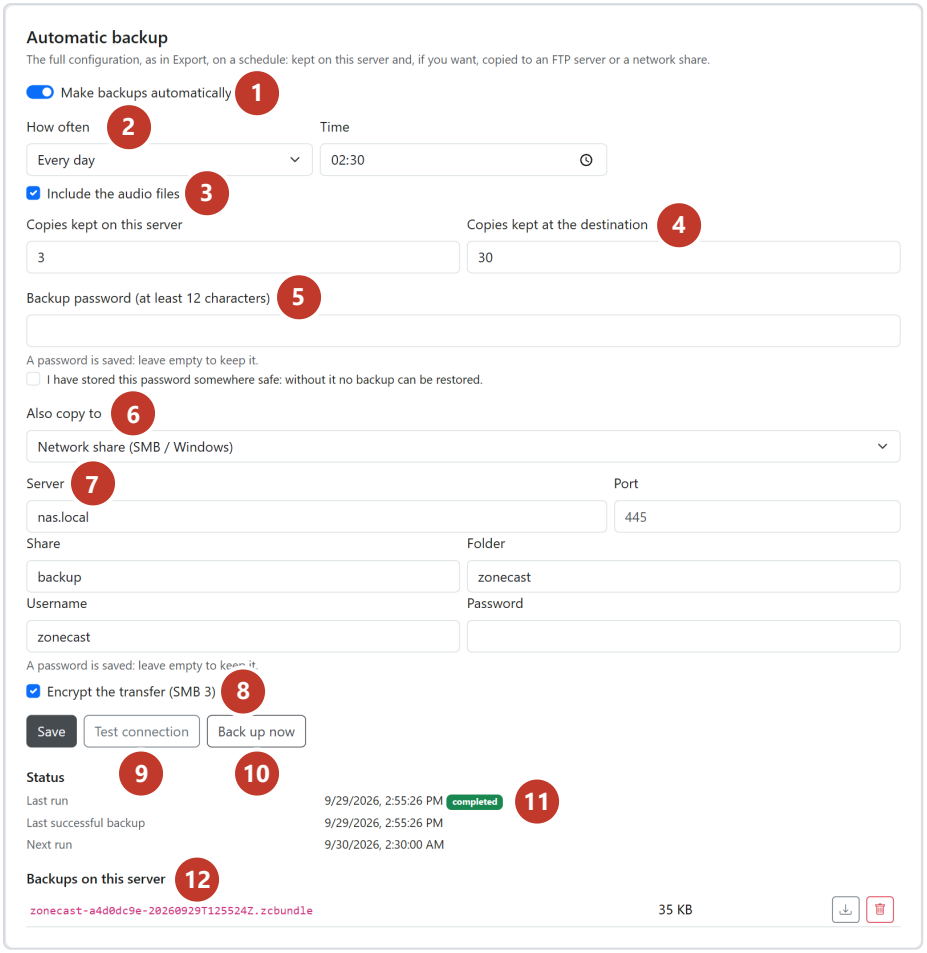
13. System — admin panel



Visible only to administrators.

1. **Network** (native install only) — the interface's IP address with subnet (e.g. 192.168.1.50/24), gateway and DNS; changes are applied at once, on trial, and must be confirmed within 2 minutes (with an on-screen countdown) by pressing **Confirm this address**; if the connection drops, reconnect at the new address and confirm from there. Otherwise they revert on their own. While a change awaits confirmation, another one can't be applied.
2. **Server time** — local time, timezone, NTP sync status, the timezone schedules run in (with a warning if it differs from the system's), NTP servers, and **Set time manually** if the server has no NTP access. NTP/timezone changes are only available on a native install (on Docker the host manages them: the panel shows the time and whether it's in sync); setting the time manually works on Docker too. On a native install, setting it manually turns NTP off until you switch it back on; moving the clock forward past a bell by more than 5 minutes makes that bell "failed".
3. **Host resources** — disk usage (with a color-coded bar warning when space is running low), RAM and CPU.
4. **Information** — installed version; click it to see the changelog (see section 15).
5. **Export full configuration** — database, media and encryption key in a single file, optionally password-protected, to transfer the installation to a new machine. It contains sensitive data (device credentials, user passwords, 2FA secrets): without a password anyone who gets it can effectively read it; the password isn't stored anywhere.
6. **Import configuration** — choose the exported file, type its password and press **Import and restart**: it replaces the current data, users and passwords included (a copy of it is still saved), and restarts the service: then sign in with the credentials of the exported installation. A file from a newer version is refused. A bell due during the restart plays as soon as the service is back (up to 5 minutes late): best to import away from bell times. This is also how you restore an automatic backup.
7. **Automatic backup** — see section 14.

14. Automatic backup

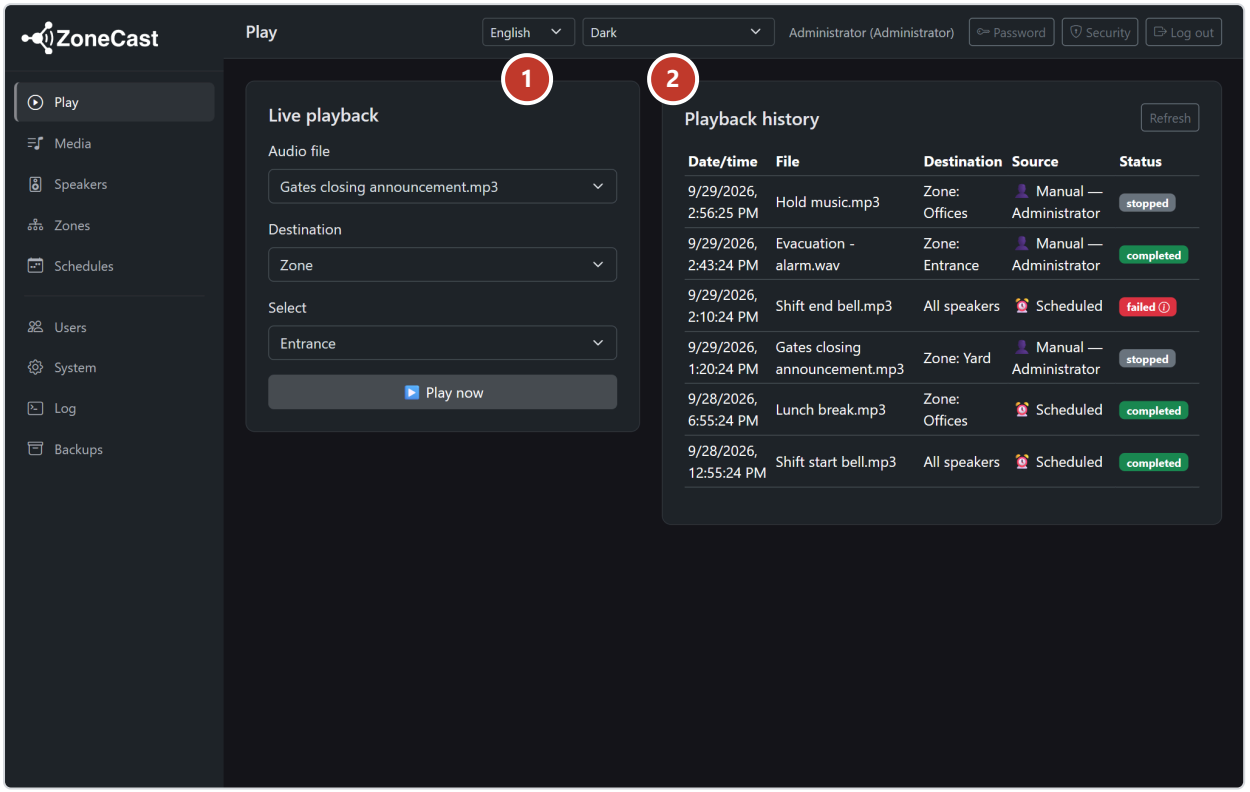


System > Automatic backup (admins only): the full configuration, as in Export, made on a schedule, kept on this server and optionally copied to an FTP server or a network share.

1. **Make backups automatically** — turns the schedule on.
2. **How often** (every day / every week, with the day) and **Time**.
3. **Include the audio files** in the backup. Without them, restored on another server the schedules have no audio (on the same server the existing files are kept).
4. **Copies kept** on this server (0 to 10) and at the destination (1 to 100, shown only when copying elsewhere); older ones made by this installation are deleted. With 0 the backup is kept only at the destination; with no other destination at least 1 copy is always kept.
5. **Backup password** (at least 12 characters) and the confirmation "*I have stored this password somewhere safe*".
6. **Also copy to** — nowhere else (this server only), FTP server with encryption (**FTPS**), network share (**SMB** / Windows), or FTP without encryption (only after ticking the consent that password and backup travel unencrypted).
7. Destination details: **Server**, **Port**, **Share** (SMB), **Folder**, **Username**, **Password** (leave empty to keep the saved one; if you change the destination type, server, port, share or username, type it again).
8. **Encrypt the transfer (SMB 3)**. For FTPS with a self-signed certificate there is a **trusted certificate fingerprint** field instead: *Test connection* shows the server's fingerprint and offers **Trust this certificate** — use it only if it matches the one shown on the server. The button only fills in the field: press **Save** to keep it.
9. **Test connection** — writes and deletes a test file at the destination.
10. **Back up now** — runs a backup immediately with the saved settings (if you changed anything, press **Save** first); it works even when automatic backups are off, as long as a backup password is saved.
11. **Status** — last run and its outcome, last successful backup, next run, and the error if the last one failed.
12. **Backups on this server** — with size, download and delete buttons.

> Without the backup password no backup can be restored. To restore, use System > **Import configuration** with the backup file and the password *it was made with*: if you later change the backup password, older backups still need the old one, so keep both. A failed or overdue backup raises a login alert; backups are paused if the settings came from another installation (after an import) until you save them again here.

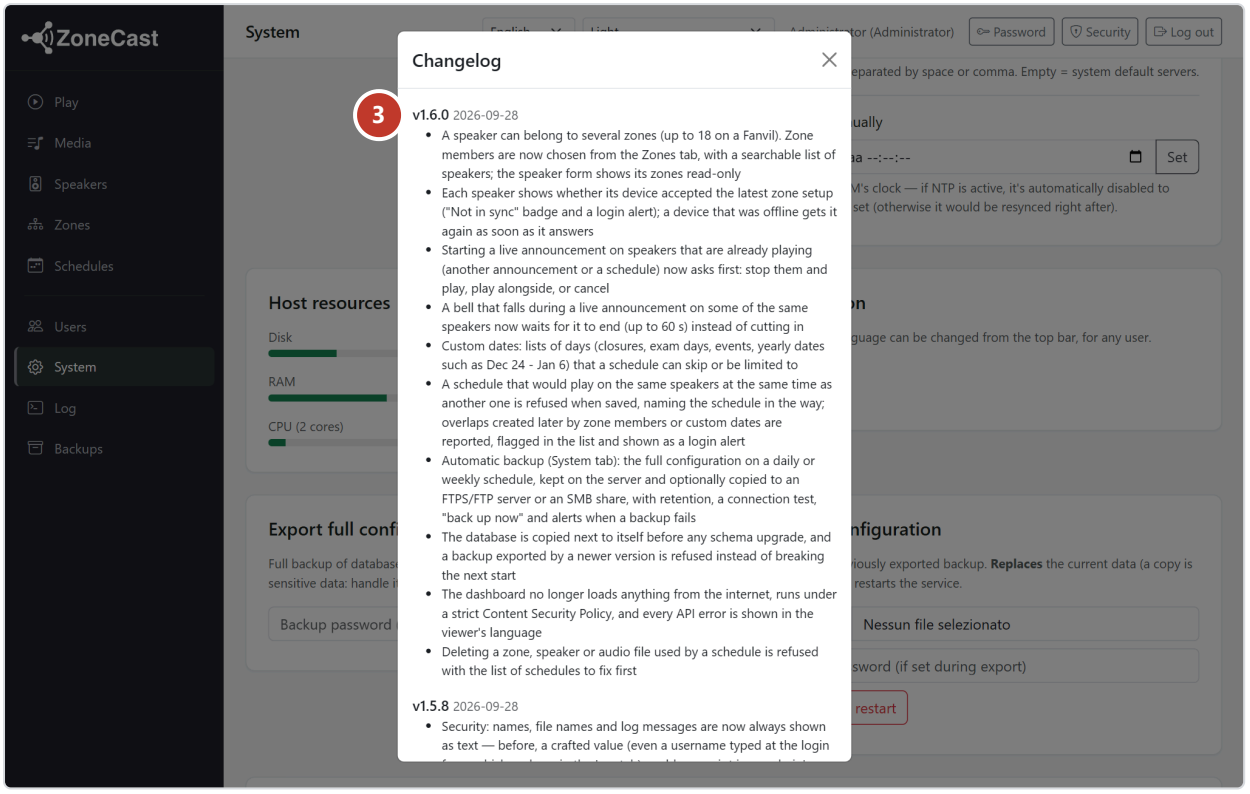
15. Language, theme, version/changelog and login alerts



Selectors available to **every** user (not just admins), at the top of the dashboard and of the login page. In the dashboard, on narrow screens (under 768 px, e.g. a phone) they move into the side menu opened with ☰, together with the Password, Security and Log out buttons.

1. **Language** — English, Italian, Chinese, Japanese, German, Spanish, French: translates the interface, including the audio-analysis tooltips and error messages. Two things are not translated: the Changelog text (always in English) and the Event log messages, which the server writes (mostly in Italian).
2. **Theme** — light, dark (shown here), or **automatic** (default): switches from light to dark following the real local sunrise/sunset time.

> Language and theme are saved in the device's browser, not in the user account: they apply to anyone using that browser. On another PC or browser they start from the defaults (English, automatic theme).

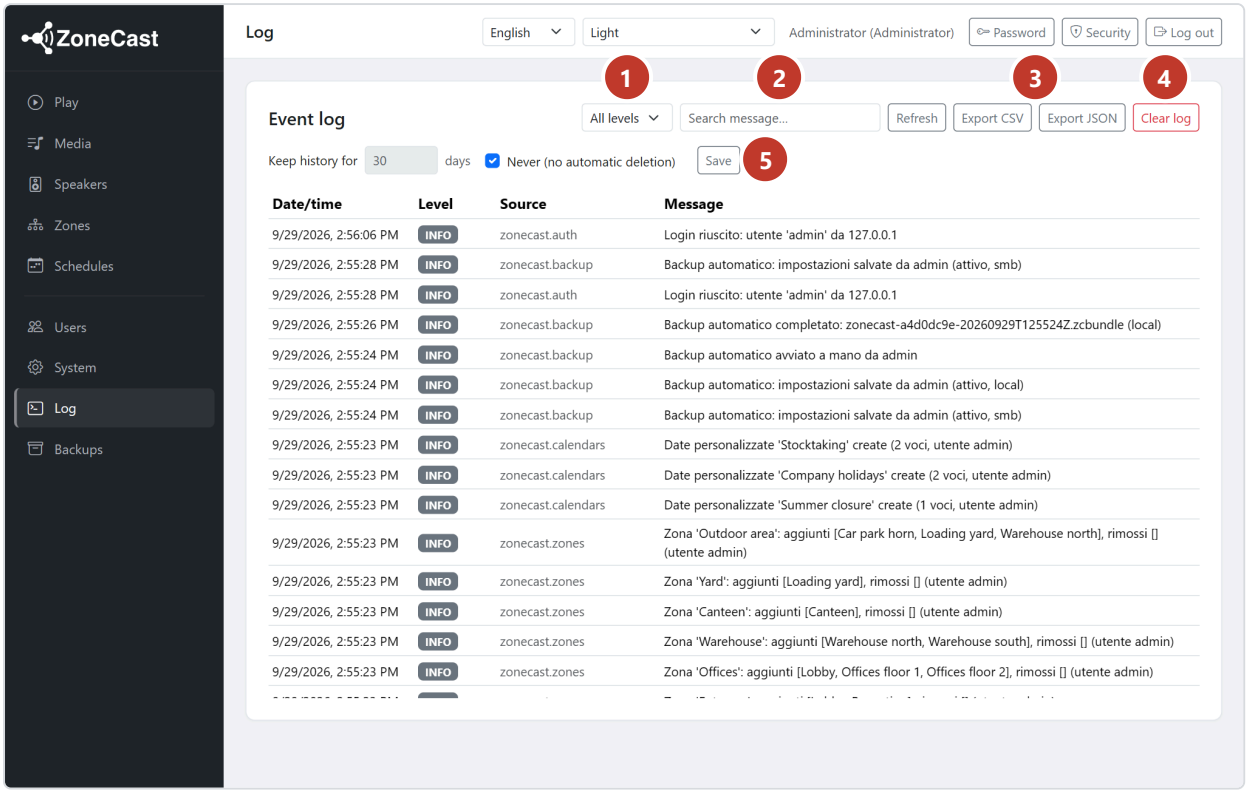


The version number in the System > **Information** card (hidden here behind the window) opens the Changelog when clicked.

3. **Changelog** — the list of changes per release, newest first: useful for quickly checking what changed after an update.

> **Login alerts** — the first time an administrator opens the dashboard in a browser session (a new tab or window), a summary appears at the top when something needs attention:
>
> - scheduled playbacks that failed in the last 24 hours;
> - disk space running low or critically low;
> - speakers that didn't accept their latest zone setup (Not in sync);
> - schedules that play on the same speakers at the same time;
> - automatic backup failed, overdue, or paused because its settings came from another installation.

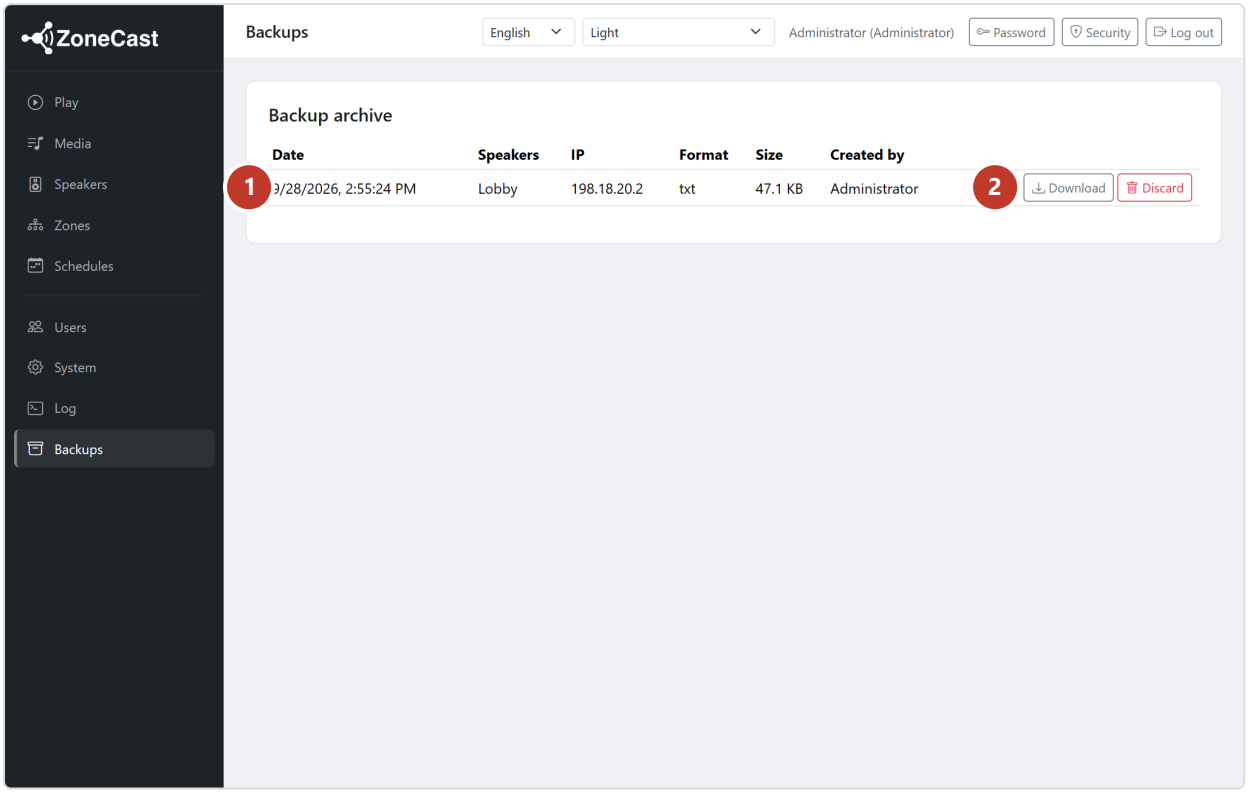
16. Event log



Persistent record of logins, device communication, playback problems, schedule runs, zone and date changes and backups — survives a service restart. Every playback is listed in Playback history (section 2). A bell skipped for a holiday, custom dates or the From/To range isn't in the history: here it leaves a line "Schedule N skipped on …" (search for "skipped"). Admins only.

1. Filter by **level** (Info/Warning/Error).
2. Search the message text.
3. **Export CSV / Export JSON** — downloads every stored event that matches the selected level and search text, not just the 300 most recent shown in the table: to export the whole history, choose all levels and clear the search first. Times in the file are in UTC.
4. **Clear log** — deletes the whole history permanently (not just the filtered rows), after asking for confirmation.
5. **Keep history for** N days (at least 1), or **Never** (no deletion by age; this is the default), then **Save**. Whatever you choose, the log keeps at most about 5000 events: beyond that the oldest are deleted automatically, so export it regularly if you need the complete history.

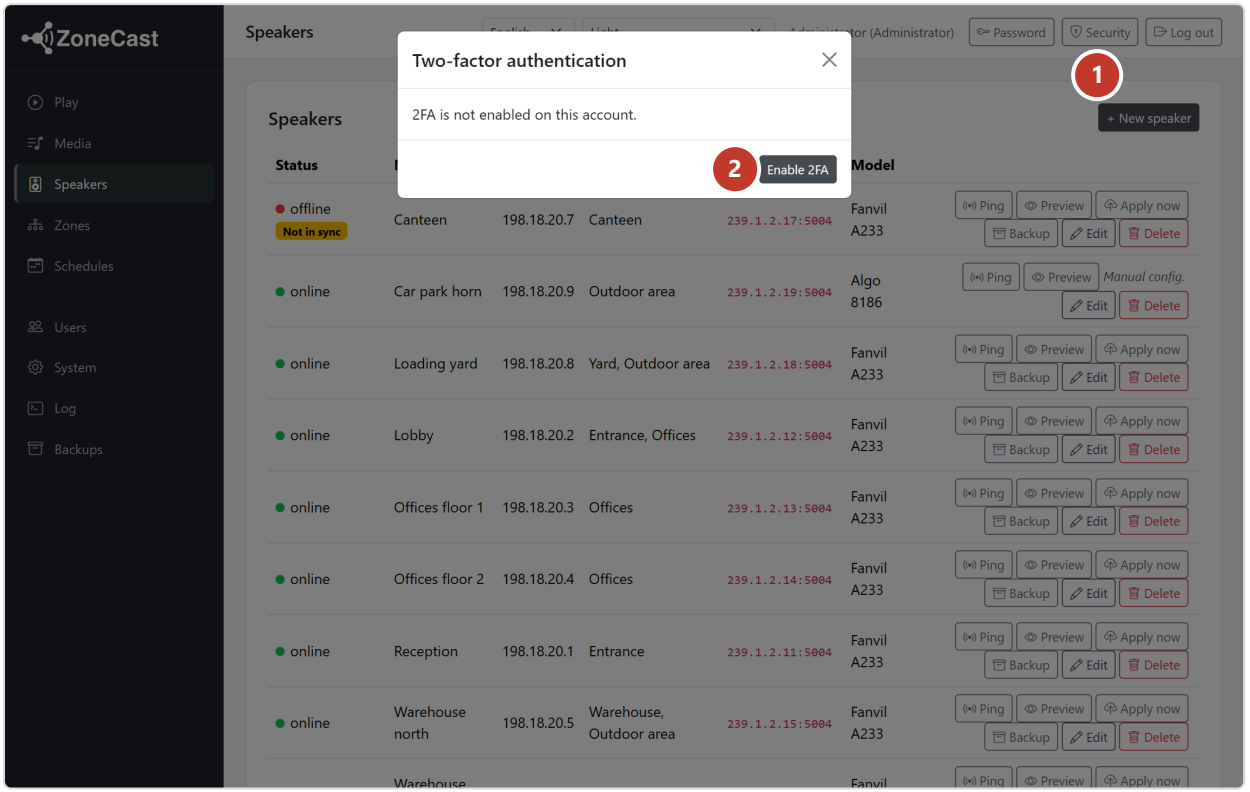
17. Backup archive



All device configuration backups (made with the **Backup** button in Speakers), including ones whose speakers have since been deleted from the app. Configuration backups of ZoneCast itself are in System > Automatic backup (section 14). Admins only.

1. Each row shows the date, the speaker's name/IP *at the time of the backup* (not its current one — so it stays identifiable even if the device was removed or reassigned), format, size and who created it. If the speaker has since been deleted, a grey **archived** badge appears next to its name.
2. **Download** or **Discard** a backup: after a confirmation it is deleted permanently.

18. Account security — 2FA



Accessible to any user, regardless of role — 2FA is optional, not enforced by the administrator.

1. **Security** button at the top (in the ☰ side menu on phones) — opens the two-factor authentication window.
2. **Enable 2FA** — shows a QR code to scan with an authenticator app (Google Authenticator, Authy, etc.); after confirming with the first 6-digit code, 10 single-use recovery codes are generated, shown only once. When 2FA is on, the same window shows **Disable 2FA** and **Regenerate recovery codes**; both ask for your password. The window shows how many recovery codes are left (each works once); regenerating them makes the old ones stop working. If you can't scan the QR, use the **Manual key**.

> Keep the recovery codes somewhere safe: they're the only way in if you lose the phone with the authenticator app. An administrator can't turn off another user's 2FA (resetting their password doesn't remove it): they can only delete the account, which is impossible for the default admin.